

Contact Center WFM Diagnostic & Executive Report

Senior Workforce Management, Operations Analytics, and Business Intelligence Assessment

Source workbook: Raw Data.xlsx | Analysis window: 2026-07-20 to 2026-07-26 | Prepared 2026-07-28

1. Executive Summary

The workbook contains a seven-day, 15-minute interval operating view with forecast volume, actual volume, AHT, service-level measures, ASA, abandonment, and staffing requirements. The main management message is **forecast volume is directionally strong in aggregate, but service performance is below target and interval-level execution is uneven**. The data supports immediate intraday WFM actions, but not yet robust seasonality claims or production-grade machine learning because the historical window is only one week and no queue, skill, agent, site, transfer, repeat-contact, shrinkage, or schedule-adherence fields are present.

Executive decision Shift the operating focus from average staffing to interval-level staffing alignment. The observed net staffing surplus of +30.2 agents coexists with a 46 interval-level understaffing events, which is consistent with timing or mix mismatch rather than a simple headcount shortage.

Evidence Service level is 72.1% versus an illustrative 80% target, ASA is 39.3 seconds, and the staffing-gap/service-level relationship is positive and statistically significant in this sample, $r=0.62$, $p<0.001$. Forecast WAPE is 4.9%, while MAPE is 13.4%, indicating that high-volume intervals are forecast reasonably well but smaller intervals have meaningful relative error.

2. Executive Dashboard

Total actual calls

Forecast WAPE

151,840

7-day observed window

4.9%

Lower is better; strong aggregate accuracy

Forecast MAPE

13.4%

Interval-level error is volatile

Service level

72.1%

Below illustrative 80% target

ASA

39.3s

Weighted average speed of answer

Abandonment

2.2%

Weighted abandonment rate

AHT

431s

P95 502s

Mean staffing gap

+30.2

Scheduled minus required agents

Data quality

98.8/100

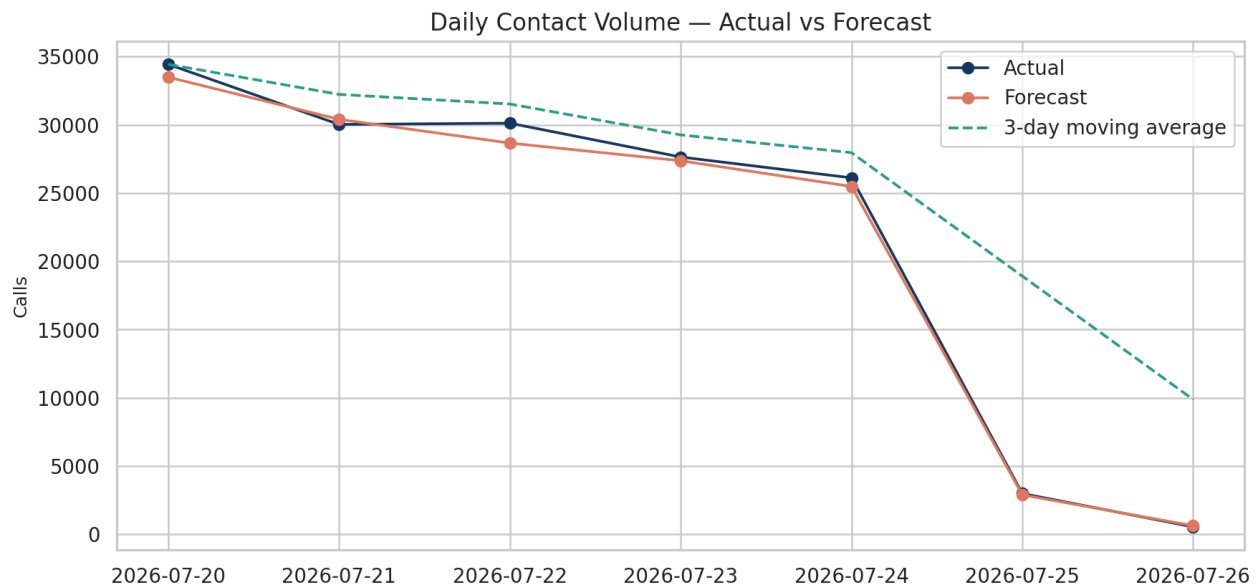
High quality, minor nulls

Overall health

6.0/10

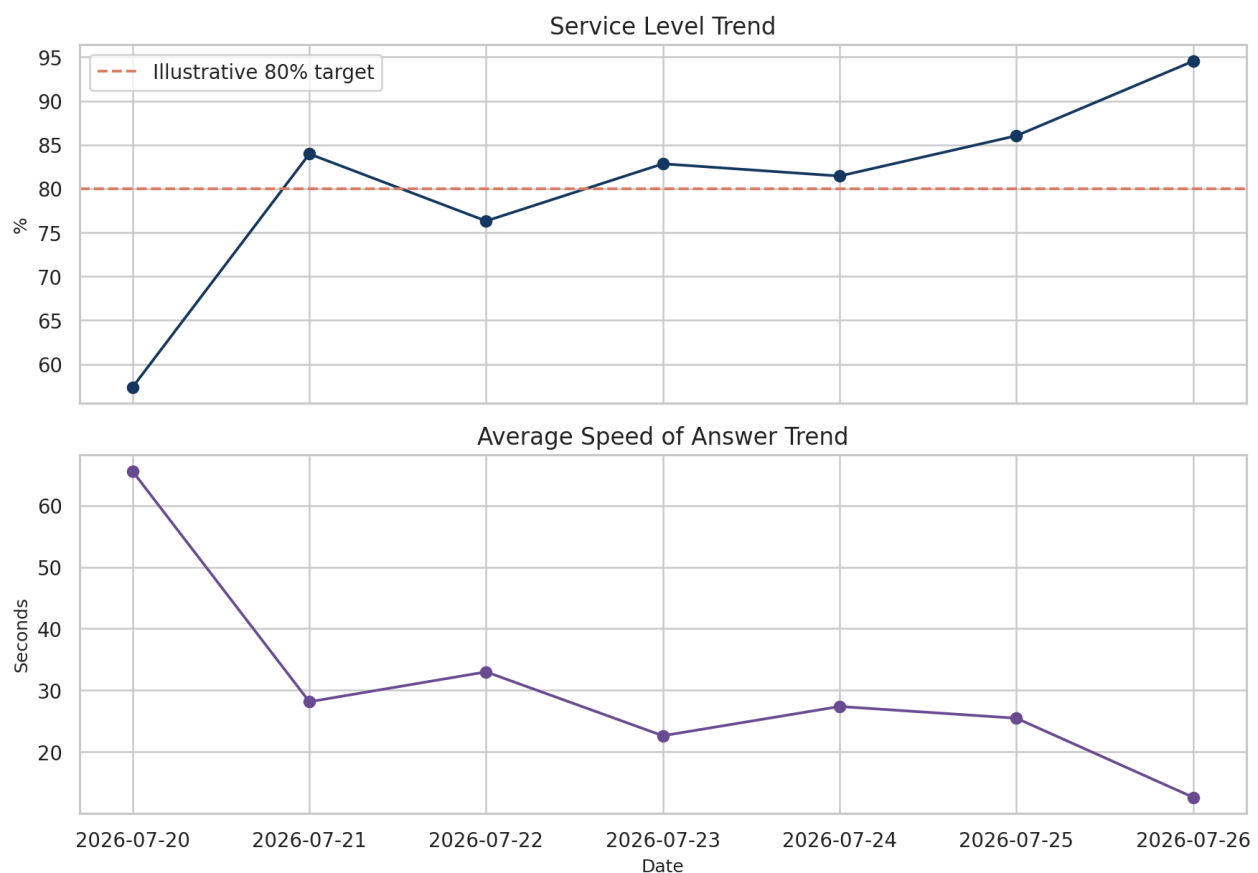
Mixed: forecast good, service risk

KPI	Current value	Benchmark / target	Variance	Risk	Interpretation
Forecast WAPE	4.9%	≤10% directional target	-5.1 pts	Low	Aggregate volume is forecastable, but interval-level errors remain material.
Forecast MAPE	13.4%	≤10% directional target	+3.4 pts	Medium	Large interval swings are masked by volume weighting.
Service level	72.1%	80% illustrative target	-7.9 pts	High	Customer wait performance is below target.
ASA	39.3s	≤30s directional target	+9.3s	Medium	Speed of answer is elevated and aligns with service-level pressure.
Abandonment	2.2%	≤3% directional target	-0.8 pts	Low/Medium	Not critical in aggregate, but interval spikes should be isolated.
Staffing gap	+30.2 agents	Near zero	+30.2	Medium	Net overstaffing exists on average, indicating timing/mix mismatch rather than pure understaffing.
Data quality	98.8/100	≥95	+3.8	Low	Suitable for directional WFM decisions with documented



Daily volume trend

Actual volume exceeded forecast on selected days, while the moving average indicates the operational baseline across the observed week.



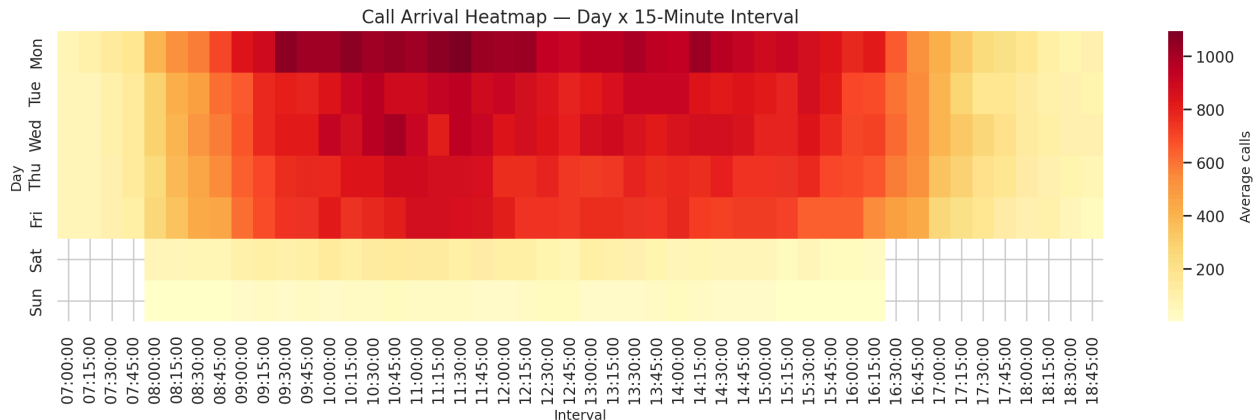
Service level and ASA trend

Service level remains below the illustrative target while ASA moves with daily operating pressure. Prioritize intervals with simultaneous low service level and high ASA.

3. Data Quality Assessment

The workbook is structurally usable for a directional WFM assessment. Completeness is high, but the source includes a small number of null records and derived/source fields that require semantic confirmation.

Data quality measure	Result
Rows analyzed	313
Columns analyzed	22
Missing cells	65 of 6,886
Completeness score	99.1%
Duplicate rows	0
Invalid dates	3
Invalid time intervals	3
Data quality score	98.8/100



Call arrival heatmap

The heatmap is the primary data-quality and operations view because it exposes missing coverage, time-granularity consistency, and demand concentration by day and interval. The visible pattern is concentrated weekday demand with a pronounced morning ramp.

Data dictionary

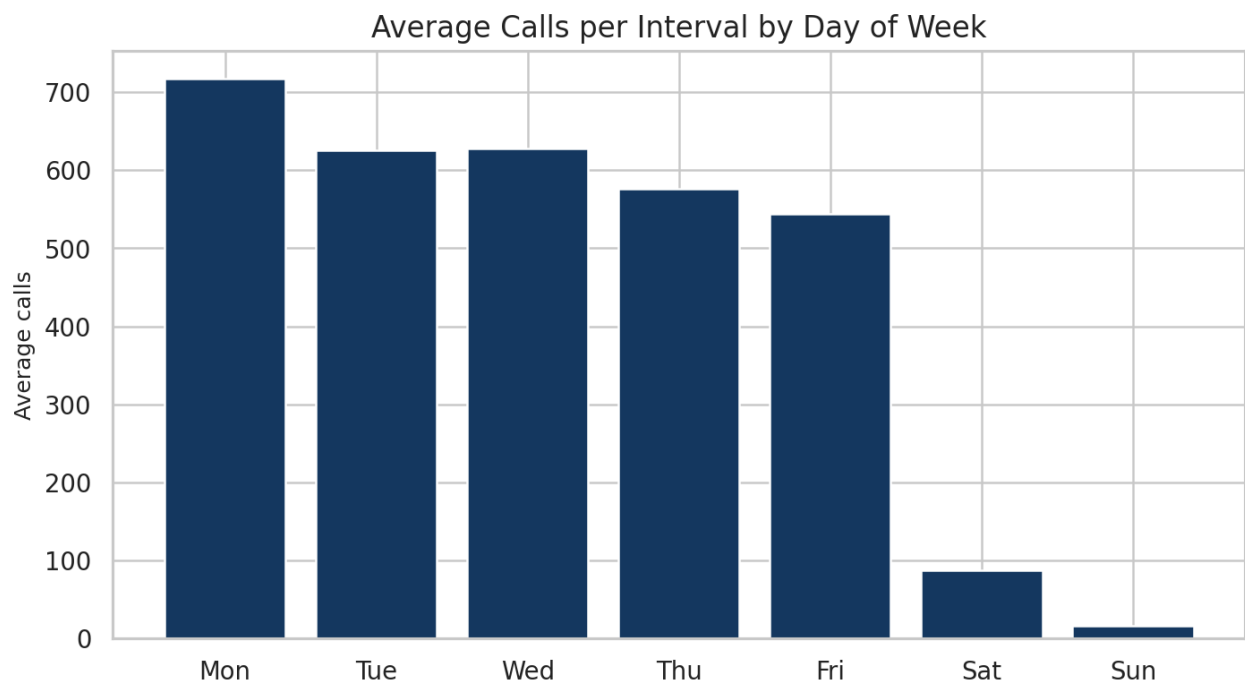
Field	Type	Unique	Nulls	Business meaning
date	datetime64[ns]	7	3	Operating date
day	object	7	3	Day-of-week label
interval	object	48	3	15-minute interval start time

interval_2	object	3	3	Interval band label
forecasted_volume	float64	311	2	Forecasted contacts
actual_volume	float64	247	3	Actual contacts
difference	float64	293	3	Source forecast error ratio
forecasted_average_handling_time_s	float64	307	2	Forecast AHT in seconds
actual_average_handling_time_s	float64	307	4	Actual AHT in seconds
workload	float64	309	3	Workload proxy, likely seconds or equivalent effort
details	object	2	3	Binary detail flag from source
difference_2	float64	289	3	Source AHT/workload difference ratio
service_level	float64	60	4	Service level ratio
fails	float64	5	0	Failure flag/count field
workload_sl	float64	301	3	Workload service-level proxy
psl	float64	284	3	Second service-level measure
abandoned_rate	float64	181	4	Abandonment ratio
average_speed_of_answer_s	float64	301	4	Average speed of answer in seconds
forecasted_agents	float64	309	3	Forecast agent requirement
required_agents	float64	308	3	Required agents
scheduled_agents	float64	273	3	Scheduled agents

Dataset relationships The workbook contains three sheets. *Raw Data* is the fact table. *Summary* and *Pivot* appear to be derivative views of the same interval-level data rather than independent sources. No stable primary key is present; a practical key would be date + interval.

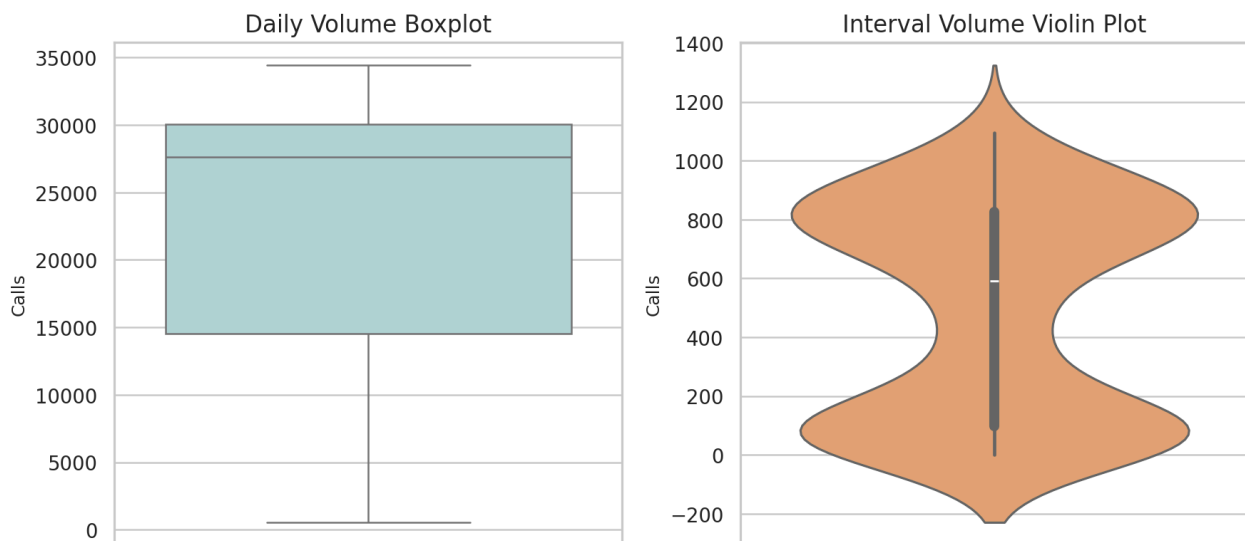
4. Call Volume Analysis

Demand is concentrated in weekday intervals, with a clear morning ramp and a broad intra-day peak. Because the dataset covers only seven dates, this section should be interpreted as an observed-week profile, not a seasonal baseline.



Average volume by day

Weekday differences are meaningful for scheduling. Use day-of-week weighting in near-term schedules, but avoid annual seasonality conclusions until at least 13 weeks are available.



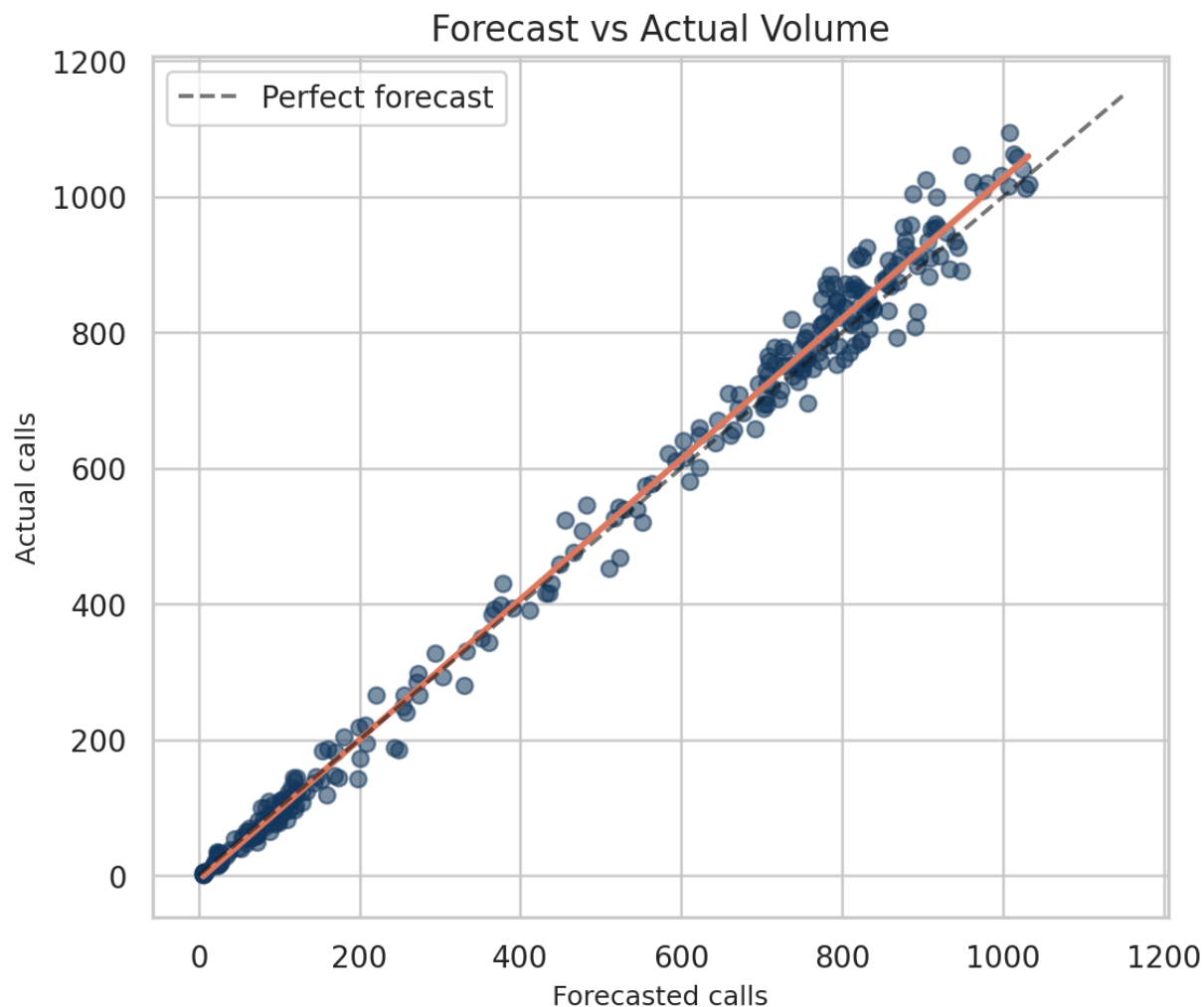
Volume distributions

The boxplot and violin plot show a wide interval-volume distribution, which increases the importance of interval-level forecasting and intraday reforecasting.

Peak interval The highest observed average interval volume is 11:30:00 at approximately 694 calls. **Trend** The seven-day sample supports a daily baseline and moving average, but not a stable growth-rate estimate.

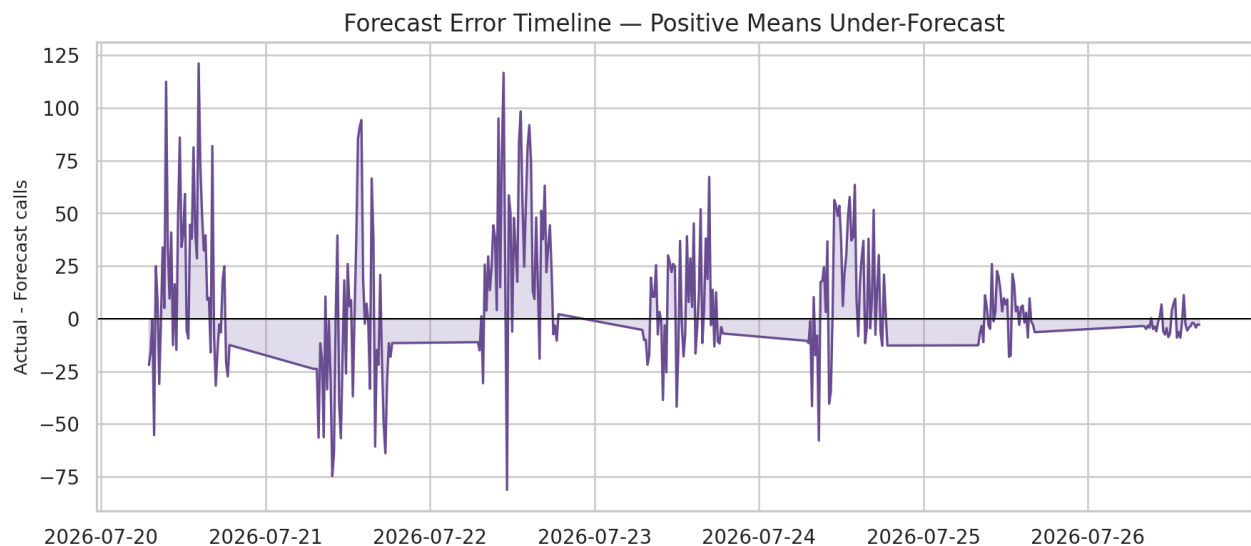
5. Forecast Assessment

Forecasting is acceptable at aggregate level but less stable at interval level. Aggregate WAPE is 4.9%, while MAPE is 13.4%. This gap is a classic signal that low-volume intervals carry high relative error even when total volume is reasonably aligned.



Forecast versus actual

Points close to the diagonal are accurate. Dispersion around the diagonal indicates interval-level error and creates staffing timing risk.



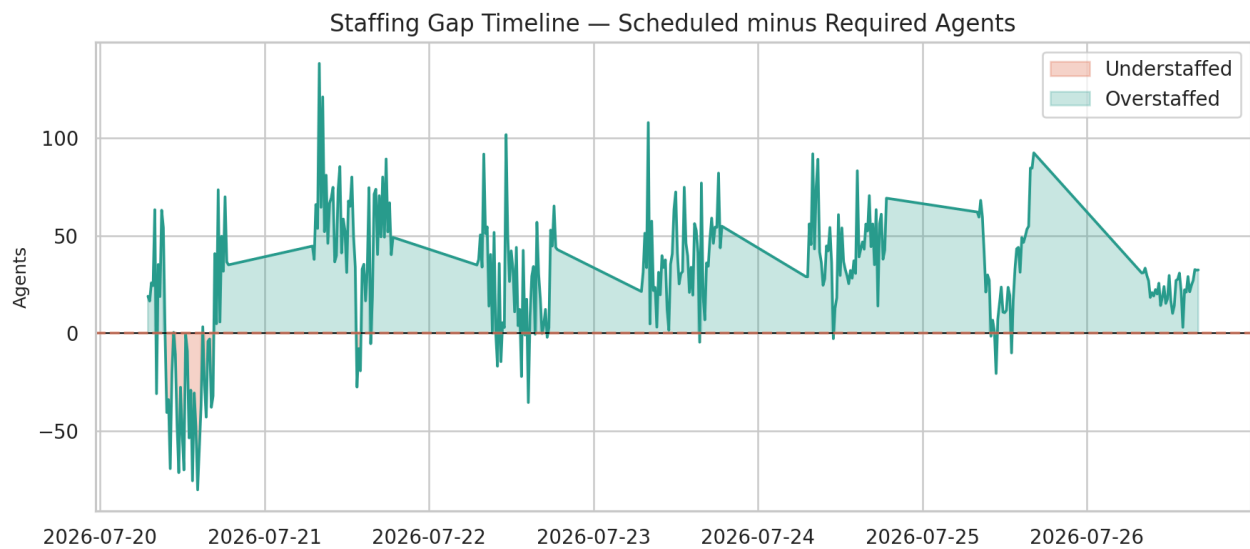
Forecast error trend

Positive error means actual demand exceeded forecast and indicates under-forecasting. Use this chart to set intraday reforecast triggers and bias corrections.

Metric	Value	Interpretation
MAE	24.2 calls	Typical absolute interval error.
RMSE	33.9 calls	Large misses are materially more costly than average misses.
Forecast bias	+1.9%	Positive bias here means actual volume exceeded forecast in aggregate.
Tracking signal	121.4	Very high because cumulative error is positive relative to MAE; investigate persistent under-forecasting or denominator sensitivity.
95% CI for mean signed interval error	-10.9% to -2.6%	Evidence of a directional error pattern in this sample.

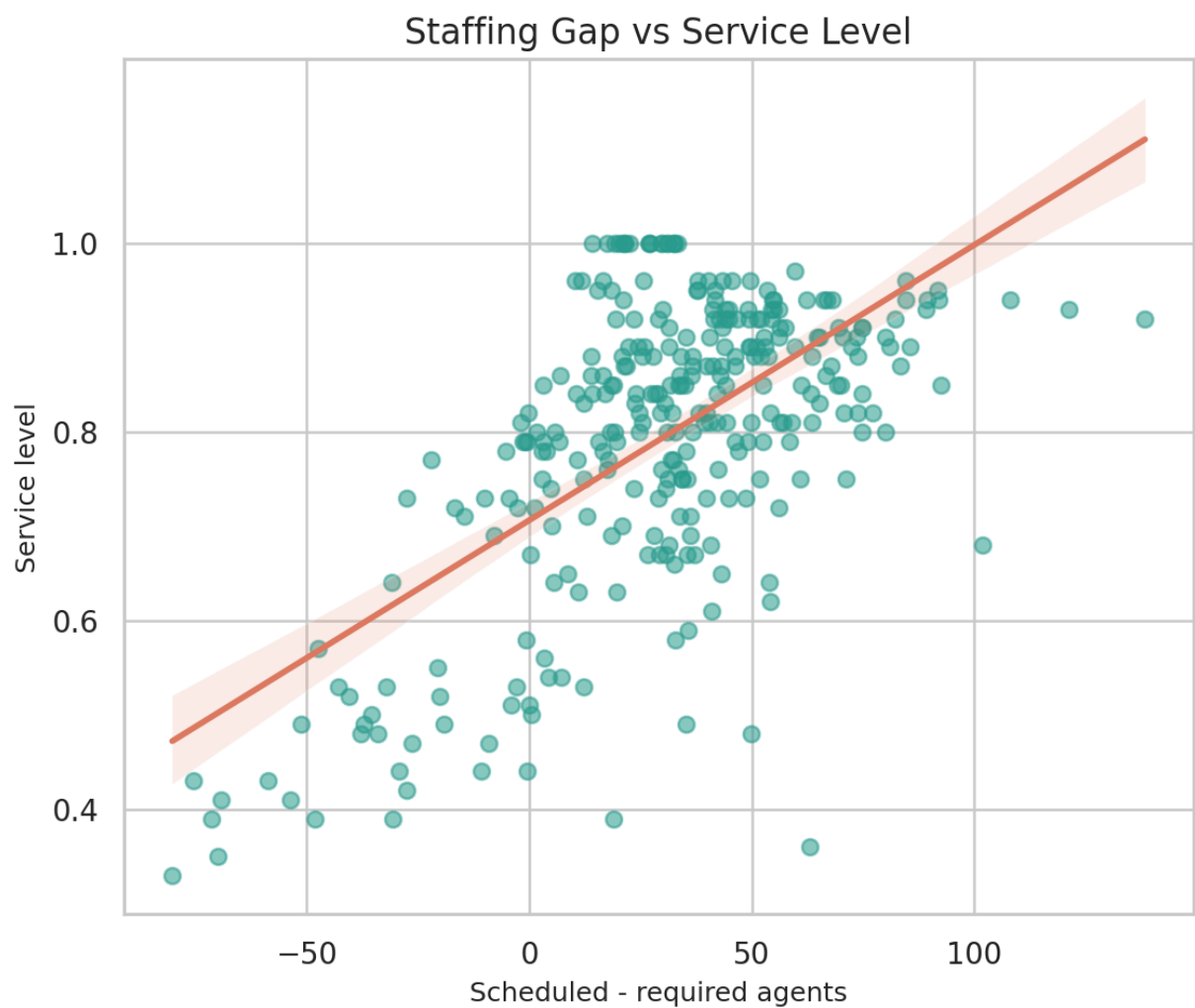
6. Staffing Analysis

The staffing picture is not a simple shortage story. Scheduled agents exceed required agents on average, yet the timeline shows understaffed intervals. This indicates that staffing is not aligned to when demand and workload occur.



Staffing gap timeline

Negative values are understaffed intervals and positive values are overstaffed intervals. The recommended action is to move capacity across intervals before adding permanent headcount.



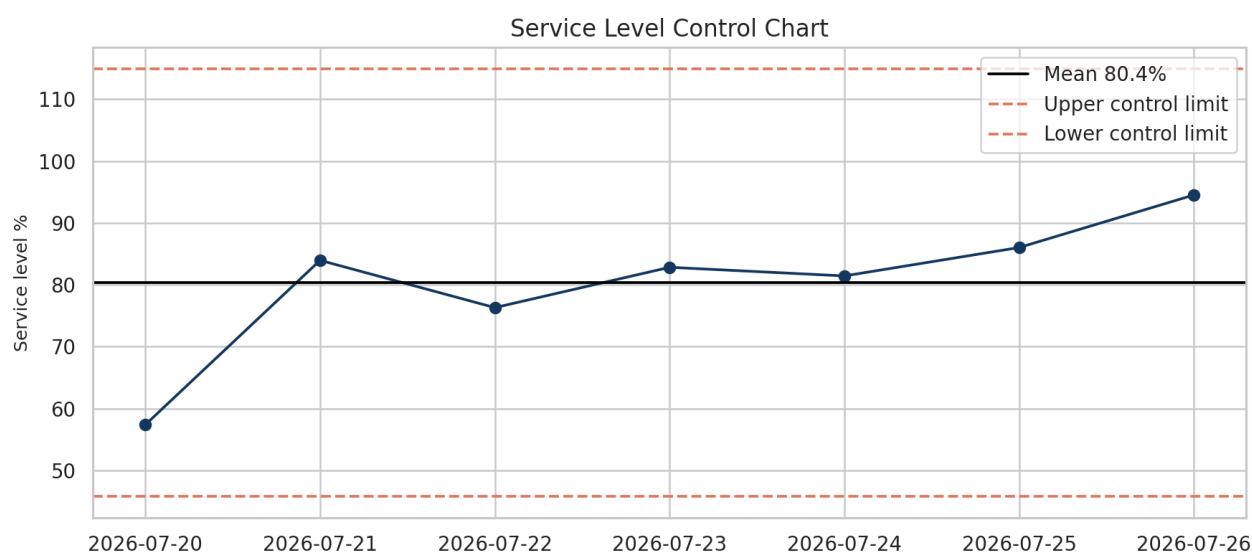
Staffing gap versus service level

The positive association supports the WFM hypothesis that additional coverage improves service level in this sample. Correlation is statistically significant, but causality is not proven by this observational dataset.

Staffing measure	Result	Business implication
Mean required agents	308.4	Base demand requirement.
Mean scheduled agents	338.7	Net scheduled capacity is above requirement.
Mean staffing gap	+30.2	Average overage, but timing mismatch exists.
Understaffed intervals	46	Immediate intraday management opportunity.
Overstaffed intervals	262	Potential redeployment or break/training placement windows.

7. Service Level Analysis

Service level is below the illustrative 80% target, with ASA elevated. The staffing/service-level relationship is statistically significant in the observed data, which supports interval rebalance, but operational leaders should validate the relationship with queue-level and interval-level controls.



Service level control chart

Control limits are directional because the sample is only seven daily observations. The chart is useful for spotting special-cause signals once the historical window is extended.

Statistical evidence $r=0.62$, $p=1.05e-34$. The 95% confidence interval for the mean interval service level is 0.8% to 0.8%.

8. Occupancy and Shrinkage Analysis

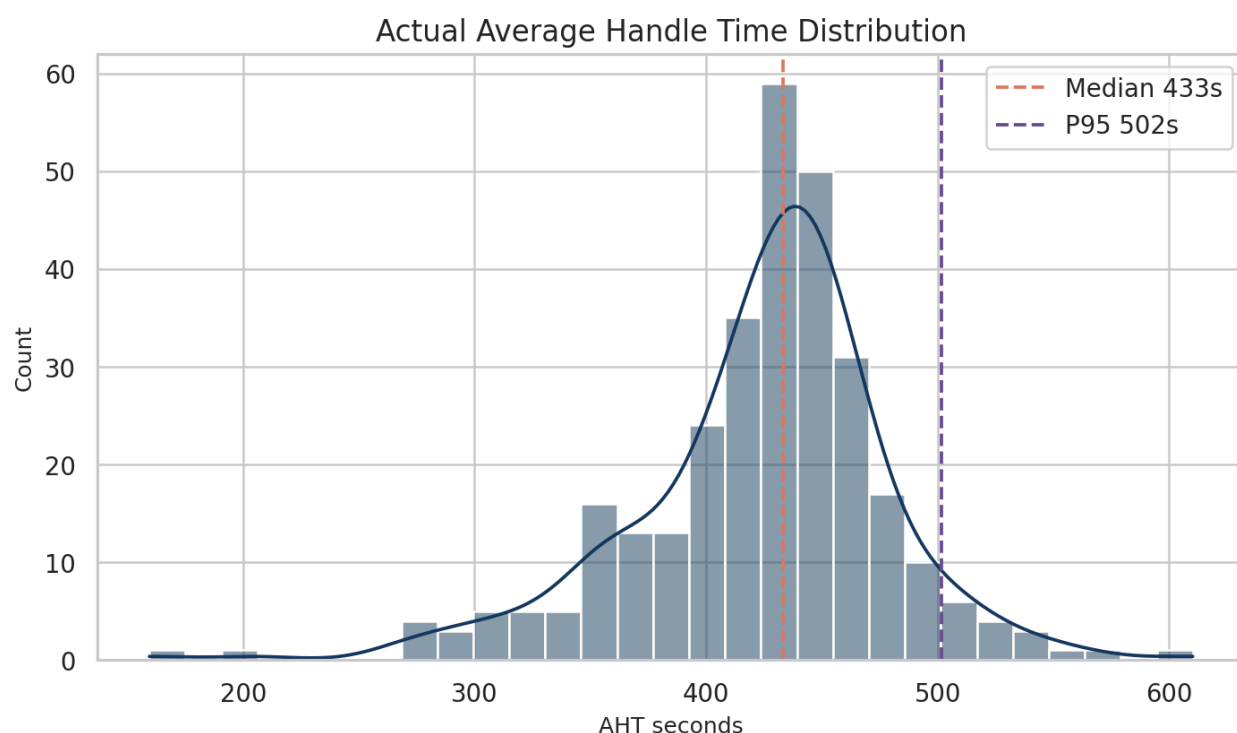
The workbook does not contain direct occupancy, shrinkage, schedule adherence, break, meeting, training, or absence fields. A workload-based occupancy proxy was computed where possible, but it should not be treated as a certified occupancy KPI.

Data gap Add agent-state and schedule-event data from NICE, Verint, Five9, Genesys, Amazon Connect, Aspect, or Calabrio to calculate true occupancy, shrinkage, adherence, utilization, and paid-hours leakage.

Recommended target ranges are 85–90% occupancy for sustainable production environments, with lower targets for complex or emotionally demanding work. The current workbook cannot determine whether teams are overloaded or idle by agent or site.

9. Handle Time Analysis

AHT is a material capacity driver. The average actual AHT is 431 seconds, with a median of 433 seconds, P90 of 478 seconds, P95 of 502 seconds, and P99 of 546 seconds. The upper tail is a likely automation and process-standardization opportunity.



AHT distribution

The right tail represents disproportionate capacity consumption. Prioritize call reasons, dispositions, and workflows associated with P95/P99 interactions once those dimensions are available.

10. Repeat Contact Analysis

Repeat-contact analysis cannot be completed from the current workbook because no customer/contact identifier, contact reason, disposition, or linkage key is available. The right next step is to add a contact-level fact table and calculate repeat within 7, 14, and 30 days.

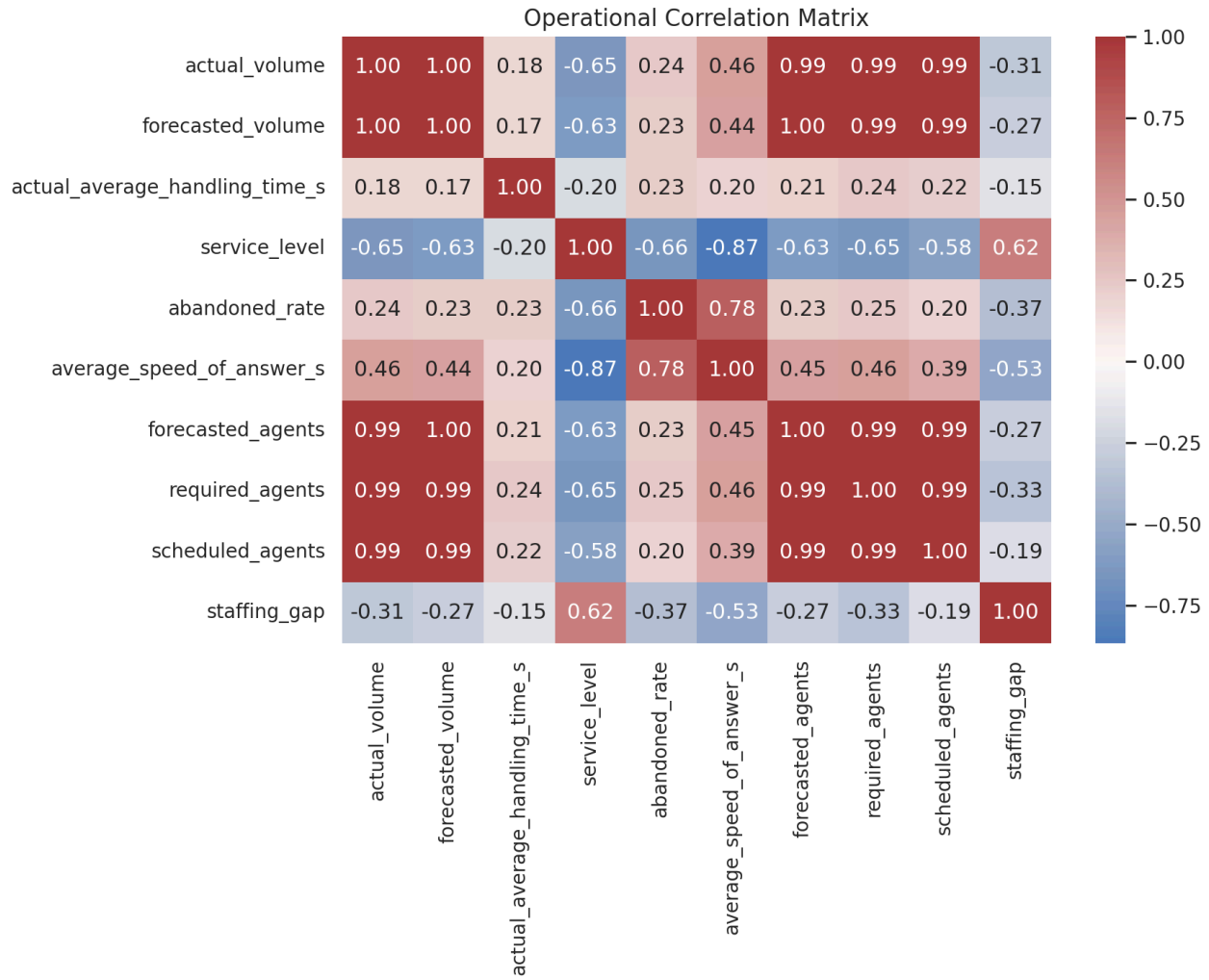
Scenario	Expected operational effect	Required data
10% repeat reduction	Reduce demand and AHT load proportionally after controlling for recontacts.	Customer/contact ID, reason, disposition, resolution code.
20% repeat reduction	Potentially material service-level improvement during peaks.	Same plus root-cause taxonomy.
30–50% repeat reduction	Strategic CX and self-service opportunity, not supportable from this workbook alone.	Validated linkage and benefit-cost model.

11. Transfer Analysis

Transfer rates, transfer chains, paths, reasons, and Sankey analysis cannot be calculated because the workbook contains no interaction-level transfer events or destination queues. A transfer event table should capture contact ID, source queue, destination queue, timestamp, reason, and handling time before and after transfer.

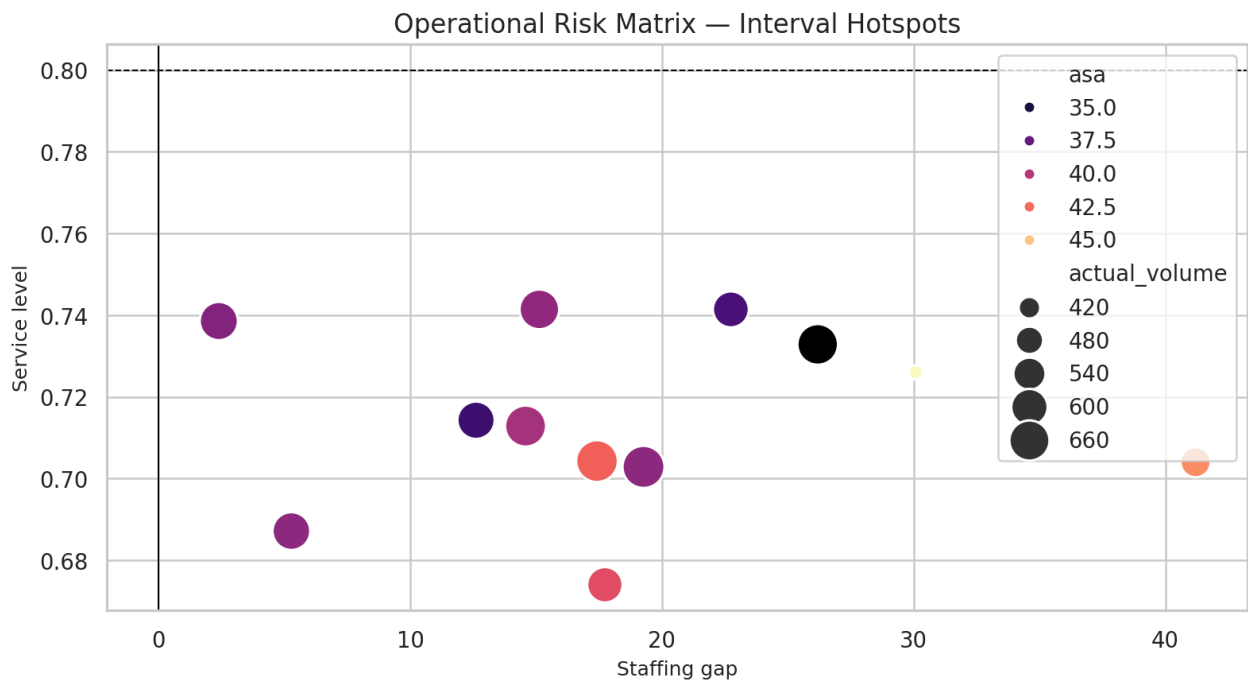
12. Statistical Analysis

The analysis includes descriptive distributions, correlation, linear regression, confidence intervals, moving averages, and unsupervised anomaly detection. Sixteen interval observations were flagged as potential anomalies using an Isolation Forest across volume, forecast error, service level, and staffing gap. These are investigation candidates, not confirmed defects.



Operational correlation matrix

The matrix highlights relationships among demand, staffing, speed, and service measures. Use it for hypothesis generation and then validate with queue, skill, and event controls.



Operational risk matrix

The upper-left area represents low staffing and lower service. These intervals should be prioritized for root-cause review and intraday playbook design.

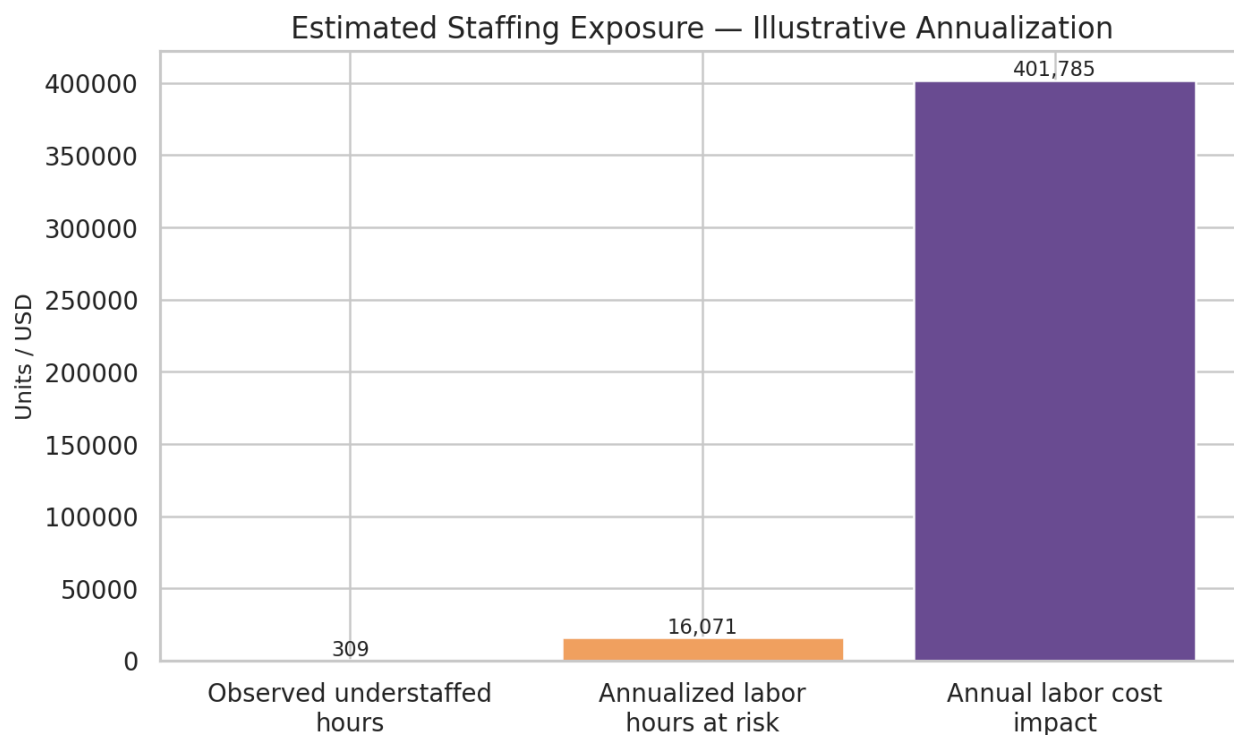
13. Machine Learning Insights

A production-grade demand model is not recommended yet because the data contains only one week of history. A small illustrative model can be fit for demonstration, but model performance would be in-sample and not decision-grade. The priority is to establish a 13–26 week interval-level data mart with holidays, promotions, queue, skill, and event features.

Recommended ML roadmap Start with seasonal naive and gradient-boosted baselines, then compare to hierarchical models by queue and interval. Use rolling-origin backtesting, WAPE/MASE, forecast bias, and service-level impact—not only R^2 .

14. Business Impact

Using an illustrative fully-loaded labor rate of \$25/hour and annualizing the observed understaffed-agent hours across 52 weeks, the current pattern implies approximately 16,071 annual labor hours of capacity at risk, or \$401,785 of annualized exposure. This is a planning estimate, not a booked financial result.

**Illustrative financial impact**

The waterfall-style view converts interval understaffing into a directional labor exposure estimate. Validate with payroll rates, productive hours, and actual service-level penalties before funding decisions.

Impact lever	Illustrative estimate	Confidence
Annual labor hours at risk	16,071	Low/medium — depends on annualization and staffing semantics.
Annual labor cost exposure	\$401,785	Low/medium — uses assumed \$25/hour loaded cost.
Potential FTE equivalent	7.7	Directional only.
Capacity improvement from 10% better interval alignment	Prioritize avoided understaffed intervals before headcount reduction.	Medium as an operating principle.

15. Risk Assessment

Rank	Risk	Level	Business impact	Mitigation
1	Service level below target	High	Customer wait and CX risk	Rebalance interval staffing and intraday actions
2	Interval forecast volatility	Medium	Staffing timing errors	Use interval-level bias correction and event flags
3	Net staffing overage masks hotspot understaffing	Medium	Paid capacity misallocated	Move from daily averages to interval heatmaps
4	Limited historical depth	High	Seasonality/model risk	Build 13-week data mart before ML deployment
5	AHT tail	Medium	Capacity consumption	Diagnose long-tail drivers and automate repeatable work

Top opportunities

Rank	Opportunity	Impact	Value hypothesis	Timing
1	Interval staffing rebalance	High	Reduce understaffed hours and improve SL	0–30 days
2	Forecast bias correction	High	Improve schedule precision	0–30 days
3	Intraday management playbook	High	Faster response to spikes	0–30 days
4	AHT tail reduction	Medium	Capacity and CX improvement	31–90 days
5	Longitudinal forecasting model	High	More defensible planning	90+ days

16. Recommendations and Executive Roadmap

Time horizon	Action	Owner	Success measure
0–30 days	Launch interval staffing heatmap and intraday reforecast trigger.	WFM Manager	Reduce understaffed intervals; improve SL and ASA.
0–30 days	Apply day/interval forecast bias correction.	Forecasting Lead	Lower WAPE and tracking signal.
31–90 days	Build root-cause taxonomy for AHT and repeat contacts.	Ops Excellence / BI	Reduce P95 AHT and recontact demand.
31–90 days	Integrate schedule events and agent states.	WFM Technology	Certified occupancy, shrinkage, adherence.
90+ days	Deploy backtested hierarchical forecast model.	Data Science / WFM	Stable forecast performance by queue and interval.
90+ days	Stand up executive dashboard with alerts and ROI tracking.	BI / Operations Strategy	Single source of truth for weekly governance.

17. Assumptions and Data Limitations

The report deliberately separates what is directly measured from what is inferred.

- The workbook covers seven observed dates from 2026-07-20 through 2026-07-26.
- Service level, abandonment, and ASA are treated as ratios/seconds as labeled in the source.
- Staffing gap is defined as scheduled agents minus required agents. Positive values indicate overage; negative values indicate understaffing.
- Industry benchmarks are directional operating targets, not a peer-adjusted benchmark study.
- Labor impact uses an illustrative \$25/hour loaded cost and annualizes one observed week to 52 weeks.
- Queue, skill, site, agent, transfer, repeat-contact, shrinkage, adherence, and schedule-event analyses require additional data fields.
- ML claims are intentionally limited because one week is insufficient for robust backtesting or seasonality modeling.

Prepared as an executive WFM diagnostic from the uploaded workbook. Recommended governance cadence — daily intraday review, weekly forecast/staffing review, monthly benefits realization review.